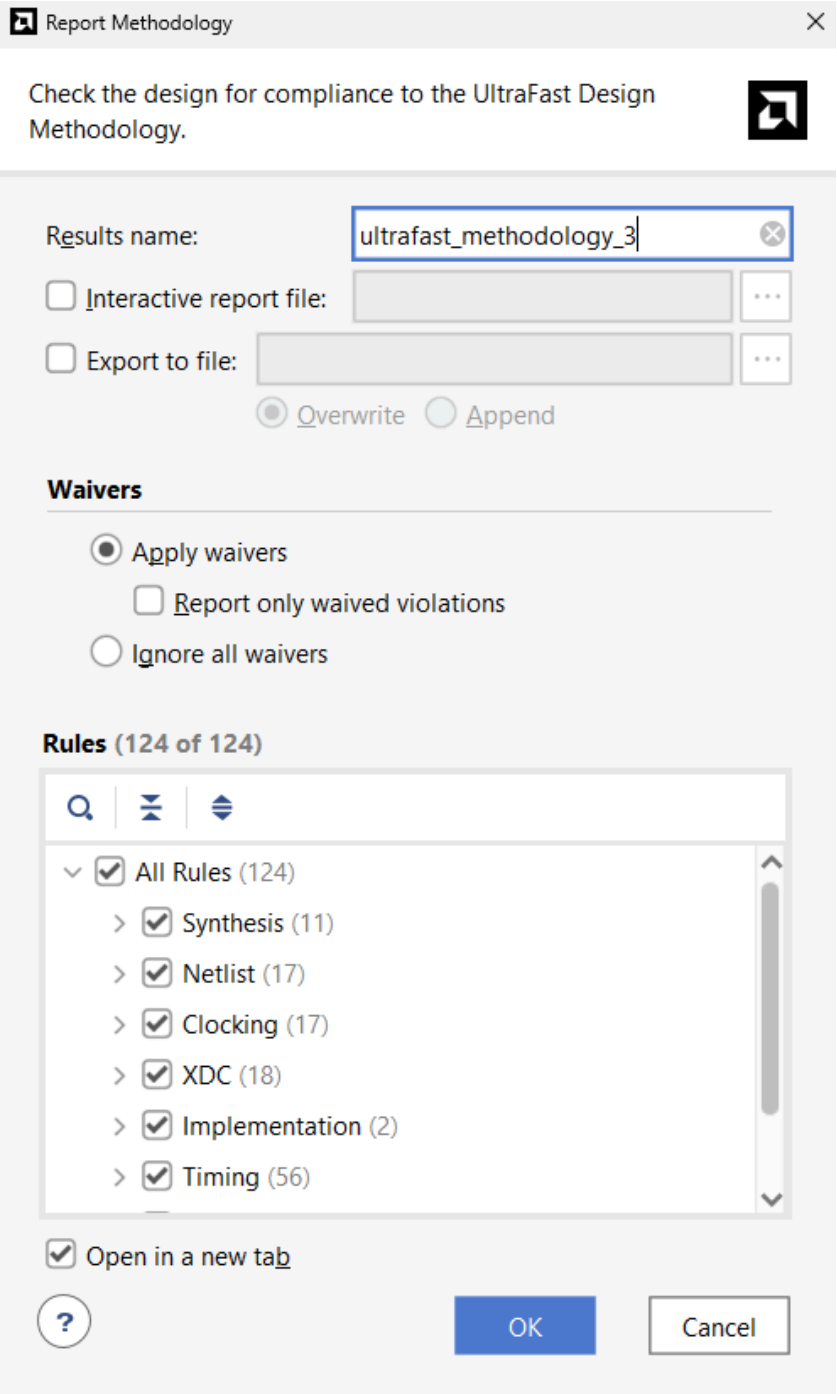
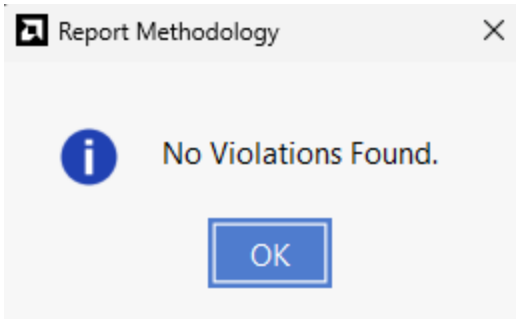
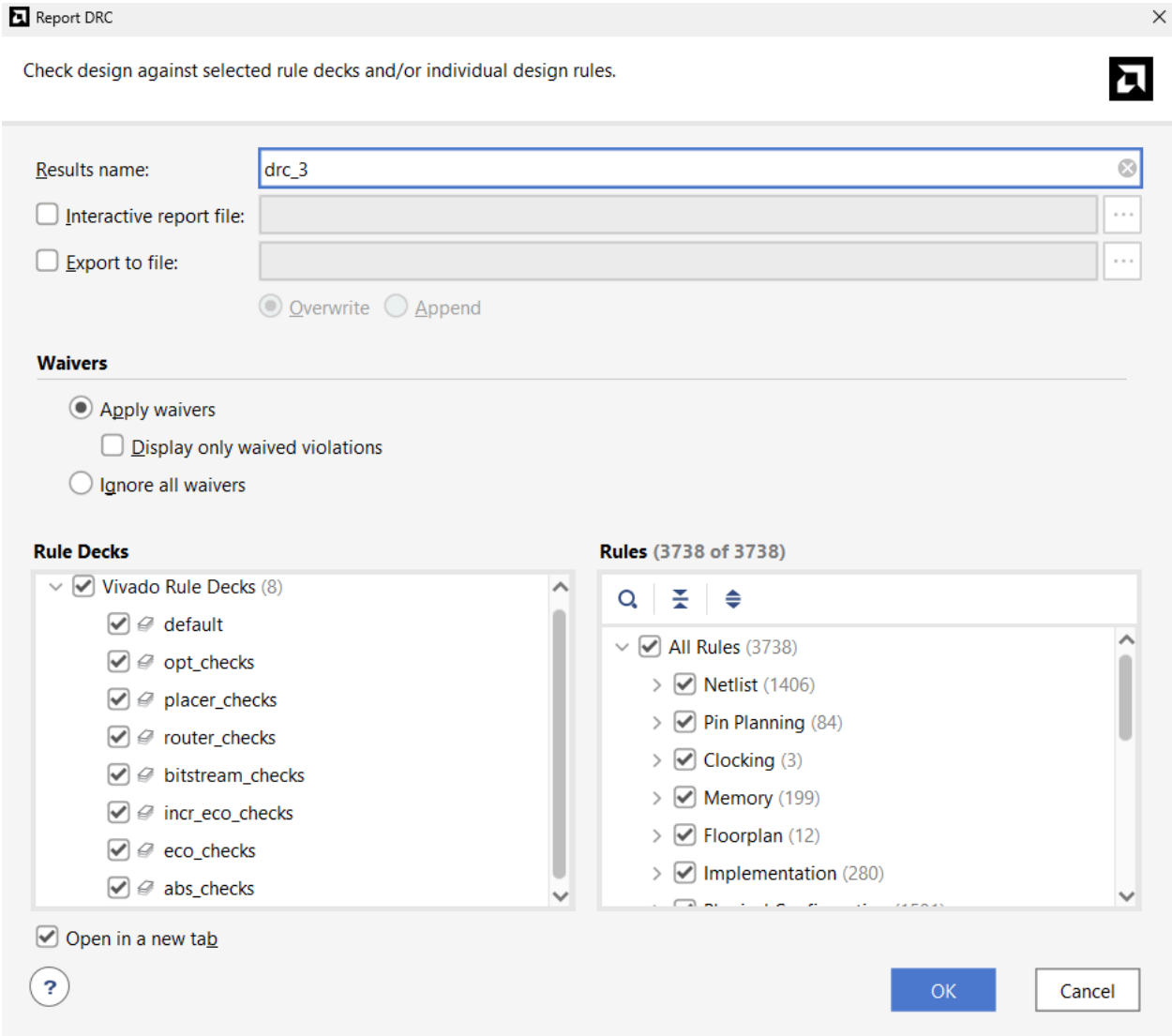
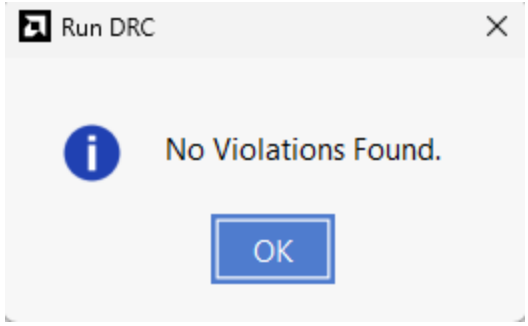
IC Design Laboratory
RISC-V Synthesis Report:




Report Methodology
No Violations Found.
OK


Report DRC
Check design against selected rule decks and/or individual design rules.
Results name:
drc_3
Interactive report file:
Export to file:
Overwrite
Append
Waivers
Apply waivers
Display only waived violations
Ignore all waivers
Rule Decks
Vivado Rule Decks (8)
default
opt_checks
placer_checks
router_checks
bitstream_checks
incr_eco_checks
eco_checks
abs_checks
Rules (3738 of 3738)
All Rules (3738)
Netlist (1406)
Pin Planning (84)
Clocking (3)
Memory (199)
Floorplan (12)
Implementation (280)
Open in a new tab
OK
Cancel



Utilization Report:

Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	F8 Muxes (8150)	Slice (8150)	LUT as Logic (20800)	LUT as Memory (9600)	Bonded IOB (106)	BUFGCTRL (32)
N basys3_top	1656	1534	320	160	863	1528	128	18	2
u_clk_divider (clock_divider)	22	29	0	0	14	22	0	0	0
u_microprocessor0 (microprocessor)	1634	1505	320	160	856	1506	128	0	0
u_core (core)	1497	1504	256	128	830	1497	0	0	0
u_decodepipeline (decode_pipe	397	191	0	0	193	397	0	0	0
u_decodestage (decode)	513	1008	256	128	576	513	0	0	0
u_executepipeline (execute_pipe	253	106	0	0	154	253	0	0	0
u_executestage (execute)	1	0	0	0	20	1	0	0	0
u_fetchpipeline (fetch_pipe)	193	63	0	0	144	193	0	0	0
u_fetchstage (fetch)	37	32	0	0	41	37	0	0	0
u_memstagepipeline (memory_p	71	104	0	0	93	71	0	0	0
u_wbstage (write_back)	32	0	0	0	29	32	0	0	0
u_data_memory (data_memory_top	136	1	64	32	39	8	128	0	0
u_memory (data_memory)	136	0	64	32	38	8	128	0	0

```
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Inc. All Rights Reserved.
------------------------------------------------------------------------------------------------
--------------------------------------------
| Tool Version : Vivado v.2024.2 (win64) Build 5239630 Fri Nov 08 22:35:27 MST 2024
| Date         : Thu Apr  2 03:15:43 2026
| Host         : Vignesh_Laptop running 64-bit major release  (build 9200)
| Command      : report_utilization -file basys3_top_utilization_placed.rpt -pb
basys3_top_utilization_placed.pb
| Design       : basys3_top
| Device       : xc7a35tcpg236-1
| Speed File   : -1
| Design State : Fully Placed
------------------------------------------------------------------------------------------------
--------------------------------------------

Utilization Design Information

Table of Contents
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1. Slice Logic
1.1 Summary of Registers by Type
2. Slice Logic Distribution
3. Memory
```

4. DSP
5. IO and GT Specific
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7. Specific Feature
8. Primitives
9. Black Boxes
10. Instantiated Netlists

1. Slice Logic

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs	1656	0	0	20800	7.96
LUT as Logic	1528	0	0	20800	7.35
LUT as Memory	128	0	0	9600	1.33
LUT as Distributed RAM	128	0			
LUT as Shift Register	0	0			
Slice Registers	1534	0	0	41600	3.69
Register as Flip Flop	1534	0	0	41600	3.69
Register as Latch	0	0	0	41600	0.00
F7 Muxes	320	0	0	16300	1.96
F8 Muxes	160	0	0	8150	1.96

* Warning! LUT value is adjusted to account for LUT combining.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-
0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
0	Yes	-	-
2	Yes	-	Set
1428	Yes	-	Reset
0	Yes	Set	-
104	Yes	Reset	-

2. Slice Logic Distribution

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice	863	0	0	8150	10.59
SLICEL	625	0			
SLICEM	238	0			
LUT as Logic	1528	0	0	20800	7.35
using O5 output only	1				
using O6 output only	1397				
using O5 and O6	130				
LUT as Memory	128	0	0	9600	1.33
LUT as Distributed RAM	128	0			

using O5 output only	0				
using O6 output only	128				
using O5 and O6	0				
LUT as Shift Register	0	0			
using O5 output only	0				
using O6 output only	0				
using O5 and O6	0				
Slice Registers	1534	0	0	41600	3.69
Register driven from within the Slice	370				
Register driven from outside the Slice	1164				
LUT in front of the register is unused	793				
LUT in front of the register is used	371				
Unique Control Sets	40		0	8150	0.49

* * Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.

3. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	50	0.00
RAMB36/FIFO*	0	0	0	50	0.00
RAMB18	0	0	0	100	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

4. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	90	0.00

5. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	18	18	0	106	16.98
IOB Master Pads	9				
IOB Slave Pads	8				
Bonded IPADs	0	0	0	10	0.00
Bonded OPADs	0	0	0	4	0.00
PHY_CONTROL	0	0	0	5	0.00
PHASER_REF	0	0	0	5	0.00
OUT_FIFO	0	0	0	20	0.00
IN_FIFO	0	0	0	20	0.00
IDELAYCTRL	0	0	0	5	0.00
IBUFDS	0	0	0	104	0.00
GTPE2_CHANNEL	0	0	0	2	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	0	20	0.00

PHASER_IN/PHASER_IN_PHY	0	0	0	20	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	0	250	0.00
IBUFDS_GTE2	0	0	0	2	0.00
ILOGIC	0	0	0	106	0.00
OLOGIC	0	0	0	106	0.00

6. Clocking

Site Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL	2	0	0	32	6.25
BUFIO	0	0	0	20	0.00
MMCME2_ADV	0	0	0	5	0.00
PLLE2_ADV	0	0	0	5	0.00
BUFMRCE	0	0	0	10	0.00
BUFHCE	0	0	0	72	0.00
BUFR	0	0	0	20	0.00

7. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%
BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
PCIE_2_1	0	0	0	1	0.00
STARTUPE2	0	0	0	1	0.00
XADC	0	0	0	1	0.00

8. Primitives

Ref Name	Used	Functional Category
FDCE	1428	Flop & Latch
LUT6	918	LUT
MUXF7	320	MuxFx
LUT5	302	LUT
LUT2	224	LUT
LUT4	161	LUT
MUXF8	160	MuxFx
RAMS64E	128	Distributed Memory
FDRE	104	Flop & Latch
LUT3	49	LUT
CARRY4	46	CarryLogic
OBUF	16	IO
LUT1	4	LUT
IBUF	2	IO

```
| FDPE     |    2 |        Flop & Latch |
| BUFG     |    2 |               Clock |
+----------+------+---------------------+


9. Black Boxes
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+----------+------+
| Ref Name | Used |
+----------+------+


10. Instantiated Netlists
-------------------------

+----------+------+
| Ref Name | Used |
+----------+------+
```

Synthesis Report:

```
#-----------------------------------------------------------
# Vivado v2024.2 (64-bit)
# SW Build 5239630 on Fri Nov 08 22:35:27 MST 2024
# IP Build 5239520 on Sun Nov 10 16:12:51 MST 2024
# SharedData Build 5239561 on Fri Nov 08 14:39:27 MST 2024
# Start of session at: Thu Apr  2 03:10:25 2026
# Process ID         : 12052
# Current directory  : C:/Users/vigne/RISCV_FinalSubmission/RISCV_FinalSubmission.runs/synth_1
# Command line       : vivado.exe -log basys3_top.vds -product Vivado -mode batch -messageDb
vivado.pb -notrace -source basys3_top.tcl
# Log file           :
C:/Users/vigne/RISCV_FinalSubmission/RISCV_FinalSubmission.runs/synth_1/basys3_top.vds
# Journal file       :
C:/Users/vigne/RISCV_FinalSubmission/RISCV_FinalSubmission.runs/synth_1\vivado.jou
# Running On         : Vignesh_Laptop
# Platform           : Windows Server 2016 or Windows 10
# Operating System   : 26200
# Processor Detail   : 12th Gen Intel(R) Core(TM) i7-1255U
# CPU Frequency      : 2611 MHz
# CPU Physical cores : 10
# CPU Logical cores  : 12
# Host memory        : 16890 MB
# Swap memory        : 15535 MB
# Total Virtual      : 32426 MB
# Available Virtual  : 2813 MB
#-----------------------------------------------------------
source basys3_top.tcl -notrace
create_project: Time (s): cpu = 00:00:06 ; elapsed = 00:00:08 . Memory (MB): peak = 366.059 ;
gain = 66.473
Command: read_checkpoint -auto_incremental -incremental
C:/Users/vigne/RISCV_FinalSubmission/RISCV_FinalSubmission.srcs/utils_1/imports/synth_1/basys3_to
p.dcp
```

```
INFO: [Vivado 12-5825] Read reference checkpoint from
C:/Users/vigne/RISCV_FinalSubmission/RISCV_FinalSubmission.srcs/utils_1/imports/synth_1/basys3_to
p.dcp for incremental synthesis
INFO: [Vivado 12-7989] Please ensure there are no constraint changes
Command: synth_design -top basys3_top -part xc7a35tcpg236-1
Starting synth_design
Attempting to get a license for feature 'Synthesis' and/or device 'xc7a35t'
INFO: [Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7a35t'
INFO: [Device 21-403] Loading part xc7a35tcpg236-1
INFO: [Designutils 20-5440] No compile time benefit to using incremental synthesis; A full
resynthesis will be run
INFO: [Designutils 20-4379] Flow is switching to default flow due to incremental criteria not
met. If you would like to alter this behaviour and have the flow terminate instead, please set
the following parameter config_implementation {autoIncr.Synth.RejectBehavior Terminate}
INFO: [Synth 8-7079] Multithreading enabled for synth_design using a maximum of 2 processes.
INFO: [Synth 8-7078] Launching helper process for spawning children vivado processes
INFO: [Synth 8-7075] Helper process launched with PID 40092
---------------------------------------------------------------------------------
Starting RTL Elaboration : Time (s): cpu = 00:00:16 ; elapsed = 00:00:18 . Memory (MB): peak =
1031.766 ; gain = 469.469
---------------------------------------------------------------------------------
INFO: [Synth 8-6157] synthesizing module 'basys3_top' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Basys3_top.v:1]
INFO: [Synth 8-6157] synthesizing module 'clock_divider' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/clock_divider.v:1]
        Parameter TOGGLE_COUNT bound to: 50000000 - type: integer
INFO: [Synth 8-6155] done synthesizing module 'clock_divider' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/clock_divider.v:1]
INFO: [Synth 8-6157] synthesizing module 'BUFG'
[C:/Vivado/2024.2/scripts/rt/data/unisim_comp.v:2676]
INFO: [Synth 8-6155] done synthesizing module 'BUFG' (0#1)
[C:/Vivado/2024.2/scripts/rt/data/unisim_comp.v:2676]
INFO: [Synth 8-6157] synthesizing module 'microprocessor' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Microprocessor.v:2]
        Parameter INSTR_MEM_FILE bound to: (null) - type: string
INFO: [Synth 8-6157] synthesizing module 'instruc_mem_top' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/instruc_mem_top.v:1]
        Parameter INIT_FILE bound to: (null) - type: string
INFO: [Synth 8-6157] synthesizing module 'instruction_memory' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/instruction_memory.v:1]
        Parameter INIT_FILE bound to: (null) - type: string
INFO: [Synth 8-6155] done synthesizing module 'instruction_memory' (0#1) [C:/Users/vigne/OneDrive
- iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/instruction_memory.v:1]
INFO: [Synth 8-6155] done synthesizing module 'instruc_mem_top' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/instruc_mem_top.v:1]
INFO: [Synth 8-6157] synthesizing module 'core' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Core.v:1]
INFO: [Synth 8-6157] synthesizing module 'fetch' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Fetch.v:1]
INFO: [Synth 8-6157] synthesizing module 'pc' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/program_counter.v:1]
INFO: [Synth 8-6155] done synthesizing module 'pc' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/program_counter.v:1]
INFO: [Synth 8-6155] done synthesizing module 'fetch' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Fetch.v:1]
INFO: [Synth 8-6157] synthesizing module 'fetch_pipe' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/fetch_pipe.v:1]
INFO: [Synth 8-6155] done synthesizing module 'fetch_pipe' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/fetch_pipe.v:1]
INFO: [Synth 8-6157] synthesizing module 'decode' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Decode.v:1]
```

```
INFO: [Synth 8-6157] synthesizing module 'controlunit' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/control_unit.v:1]
INFO: [Synth 8-6157] synthesizing module 'type_decoder' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/type_decoder.v:3]
INFO: [Synth 8-6155] done synthesizing module 'type_decoder' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/type_decoder.v:3]
INFO: [Synth 8-6157] synthesizing module 'control_decoder' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/control_decoder.v:1]
INFO: [Synth 8-6155] done synthesizing module 'control_decoder' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/control_decoder.v:1]
INFO: [Synth 8-6155] done synthesizing module 'controlunit' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/control_unit.v:1]
INFO: [Synth 8-6157] synthesizing module 'immediategen' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/immediate_gen.v:1]
INFO: [Synth 8-6155] done synthesizing module 'immediategen' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/immediate_gen.v:1]
INFO: [Synth 8-6157] synthesizing module 'registerfile' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/register_file.v:1]
INFO: [Synth 8-6155] done synthesizing module 'registerfile' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/register_file.v:1]
INFO: [Synth 8-6157] synthesizing module 'mux' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/mux1_2.v:1]
INFO: [Synth 8-6155] done synthesizing module 'mux' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/mux1_2.v:1]
INFO: [Synth 8-6155] done synthesizing module 'decode' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Decode.v:1]
INFO: [Synth 8-6157] synthesizing module 'decode_pipe' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/decode_pipe.v:1]
INFO: [Synth 8-6155] done synthesizing module 'decode_pipe' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/decode_pipe.v:1]
INFO: [Synth 8-6157] synthesizing module 'execute_forwarding' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/execute_forwarding.v:1]
INFO: [Synth 8-6155] done synthesizing module 'execute_forwarding' (0#1) [C:/Users/vigne/OneDrive
- iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/execute_forwarding.v:1]
INFO: [Synth 8-6157] synthesizing module 'branch' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/branch.v:1]
INFO: [Synth 8-6155] done synthesizing module 'branch' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/branch.v:1]
INFO: [Synth 8-6157] synthesizing module 'execute' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Execute.v:1]
INFO: [Synth 8-6157] synthesizing module 'alu' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/alu.v:1]
INFO: [Synth 8-6155] done synthesizing module 'alu' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/alu.v:1]
INFO: [Synth 8-6157] synthesizing module 'adder' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/adder.v:1]
INFO: [Synth 8-6155] done synthesizing module 'adder' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/adder.v:1]
INFO: [Synth 8-6155] done synthesizing module 'execute' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Execute.v:1]
INFO: [Synth 8-6157] synthesizing module 'execute_pipe' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/execute_pipe.v:1]
INFO: [Synth 8-6155] done synthesizing module 'execute_pipe' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/execute_pipe.v:1]
INFO: [Synth 8-6157] synthesizing module 'memory_stage' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/memory_stage.v:1]
INFO: [Synth 8-6157] synthesizing module 'wrappermem' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/wrapper_memory.v:1]
INFO: [Synth 8-226] default block is never used [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/wrapper_memory.v:23]
INFO: [Synth 8-226] default block is never used [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/wrapper_memory.v:76]
```

```
INFO: [Synth 8-226] default block is never used [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/wrapper_memory.v:117]
INFO: [Synth 8-6155] done synthesizing module 'wrappermem' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/wrapper_memory.v:1]
INFO: [Synth 8-6155] done synthesizing module 'memory_stage' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/memory_stage.v:1]
INFO: [Synth 8-6157] synthesizing module 'memory_pipe' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/memstage_pipe.v:1]
INFO: [Synth 8-6155] done synthesizing module 'memory_pipe' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/memstage_pipe.v:1]
INFO: [Synth 8-6157] synthesizing module 'write_back' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Write_back.v:1]
INFO: [Synth 8-6157] synthesizing module 'mux2_4' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/mux2_4.v:1]
INFO: [Synth 8-226] default block is never used [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/mux2_4.v:8]
INFO: [Synth 8-6155] done synthesizing module 'mux2_4' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/mux2_4.v:1]
INFO: [Synth 8-6155] done synthesizing module 'write_back' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Write_back.v:1]
INFO: [Synth 8-6155] done synthesizing module 'core' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Core.v:1]
INFO: [Synth 8-6157] synthesizing module 'data_memory_top' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/data_memory_top.v:1]
INFO: [Synth 8-6157] synthesizing module 'data_memory' [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/data_memory.v:1]
INFO: [Synth 8-6155] done synthesizing module 'data_memory' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/data_memory.v:1]
INFO: [Synth 8-6155] done synthesizing module 'data_memory_top' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/data_memory_top.v:1]
INFO: [Synth 8-6155] done synthesizing module 'microprocessor' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Microprocessor.v:2]
INFO: [Synth 8-6155] done synthesizing module 'basys3_top' (0#1) [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/Basys3_top.v:1]
WARNING: [Synth 8-7129] Port alu_out_address[31] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[30] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[29] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[28] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[27] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[26] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[25] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[24] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[23] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[22] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[21] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[20] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[19] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[18] in module memory_stage is either unconnected or
has no load
```

```
WARNING: [Synth 8-7129] Port alu_out_address[17] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[16] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[15] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[14] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[13] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[12] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[11] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[10] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[9] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[8] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[7] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[6] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[5] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[4] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[3] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port alu_out_address[2] in module memory_stage is either unconnected or
has no load
WARNING: [Synth 8-7129] Port instruction[31] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[30] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[29] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[28] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[27] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[26] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[25] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[24] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[23] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[22] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[21] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[20] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[19] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[18] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[17] in module memory_stage is either unconnected or has
no load
```

```
WARNING: [Synth 8-7129] Port instruction[16] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[15] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[11] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[10] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[9] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[8] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[7] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[6] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[5] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[4] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[3] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[2] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[1] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction[0] in module memory_stage is either unconnected or has
no load
WARNING: [Synth 8-7129] Port instruction_execute[31] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[30] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[29] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[28] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[27] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[26] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[25] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[24] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[23] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[22] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[21] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[20] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[19] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[18] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[17] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[16] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[15] in module execute_forwarding is either
unconnected or has no load
```

```
WARNING: [Synth 8-7129] Port instruction_execute[14] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[13] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[12] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[11] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[10] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[9] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[8] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instruction_execute[7] in module execute_forwarding is either
unconnected or has no load
WARNING: [Synth 8-7129] Port instr[6] in module immediategen is either unconnected or has no load
WARNING: [Synth 8-7129] Port instr[5] in module immediategen is either unconnected or has no load
WARNING: [Synth 8-7129] Port instr[4] in module immediategen is either unconnected or has no load
WARNING: [Synth 8-7129] Port instr[3] in module immediategen is either unconnected or has no load
WARNING: [Synth 8-7129] Port instr[2] in module immediategen is either unconnected or has no load
WARNING: [Synth 8-7129] Port instr[1] in module immediategen is either unconnected or has no load
WARNING: [Synth 8-7129] Port instr[0] in module immediategen is either unconnected or has no load
WARNING: [Synth 8-7129] Port valid in module type_decoder is either unconnected or has no load
WARNING: [Synth 8-7129] Port load_signal_controller in module type_decoder is either unconnected
or has no load
WARNING: [Synth 8-7129] Port instruction_rd[31] in module decode is either unconnected or has no
load
WARNING: [Synth 8-7129] Port instruction_rd[30] in module decode is either unconnected or has no
load
WARNING: [Synth 8-7129] Port instruction_rd[29] in module decode is either unconnected or has no
load
WARNING: [Synth 8-7129] Port instruction_rd[28] in module decode is either unconnected or has no
load
WARNING: [Synth 8-7129] Port instruction_rd[27] in module decode is either unconnected or has no
load
WARNING: [Synth 8-7129] Port instruction_rd[26] in module decode is either unconnected or has no
load
WARNING: [Synth 8-7129] Port instruction_rd[25] in module decode is either unconnected or has no
load
INFO: [Common 17-14] Message 'Synth 8-7129' appears 100 times and further instances of the
messages will be disabled. Use the Tcl command set_msg_config to change the current settings.
---------------------------------------------------------------------------------
Finished RTL Elaboration : Time (s): cpu = 00:00:21 ; elapsed = 00:00:24 . Memory (MB): peak =
1146.734 ; gain = 584.438
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Start Handling Custom Attributes
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Finished Handling Custom Attributes : Time (s): cpu = 00:00:22 ; elapsed = 00:00:24 . Memory
(MB): peak = 1146.734 ; gain = 584.438
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Finished RTL Optimization Phase 1 : Time (s): cpu = 00:00:22 ; elapsed = 00:00:24 . Memory (MB):
peak = 1146.734 ; gain = 584.438
---------------------------------------------------------------------------------
Netlist sorting complete. Time (s): cpu = 00:00:01 ; elapsed = 00:00:00.155 . Memory (MB): peak =
1146.734 ; gain = 0.000
INFO: [Project 1-570] Preparing netlist for logic optimization

Processing XDC Constraints
```

```
Initializing timing engine
Parsing XDC File [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/basys3_top.xdc]
Finished Parsing XDC File [C:/Users/vigne/OneDrive -
iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/basys3_top.xdc]
INFO: [Project 1-236] Implementation specific constraints were found while reading constraint
file [C:/Users/vigne/OneDrive - iitgn.ac.in/Desktop/FPGA_f/FPGA_f/design/basys3_top.xdc]. These
constraints will be ignored for synthesis but will be used in implementation. Impacted
constraints are listed in the file [.Xil/basys3_top_propImpl.xdc].
Resolution: To avoid this warning, move constraints listed in [.Xil/basys3_top_propImpl.xdc] to
another XDC file and exclude this new file from synthesis with the used_in_synthesis property
(File Properties dialog in GUI) and re-run elaboration/synthesis.
Completed Processing XDC Constraints

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.001 . Memory (MB): peak =
1235.371 ; gain = 0.000
INFO: [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed.

Constraint Validation Runtime : Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.047 . Memory (MB):
peak = 1235.371 ; gain = 0.000
INFO: [Designutils 20-5440] No compile time benefit to using incremental synthesis; A full
resynthesis will be run
INFO: [Designutils 20-4379] Flow is switching to default flow due to incremental criteria not
met. If you would like to alter this behaviour and have the flow terminate instead, please set
the following parameter config_implementation {autoIncr.Synth.RejectBehavior Terminate}
---------------------------------------------------------------------------------
Finished Constraint Validation : Time (s): cpu = 00:00:44 ; elapsed = 00:00:49 . Memory (MB):
peak = 1235.371 ; gain = 673.074
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Start Loading Part and Timing Information
---------------------------------------------------------------------------------
Loading part: xc7a35tcpg236-1
---------------------------------------------------------------------------------
Finished Loading Part and Timing Information : Time (s): cpu = 00:00:44 ; elapsed = 00:00:49 .
Memory (MB): peak = 1235.371 ; gain = 673.074
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Start Applying 'set_property' XDC Constraints
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Finished applying 'set_property' XDC Constraints : Time (s): cpu = 00:00:44 ; elapsed = 00:00:49
. Memory (MB): peak = 1235.371 ; gain = 673.074
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Finished RTL Optimization Phase 2 : Time (s): cpu = 00:00:46 ; elapsed = 00:00:52 . Memory (MB):
peak = 1235.371 ; gain = 673.074
---------------------------------------------------------------------------------
---------------------------------------------------------------------------------
Start RTL Component Statistics
---------------------------------------------------------------------------------
Detailed RTL Component Info :
+---Adders :
          2 Input   32 Bit       Adders := 3
          3 Input   32 Bit       Adders := 1
          2 Input   28 Bit       Adders := 1
+---XORs :
          2 Input     32 Bit         XORs := 1
+---Registers :
                      32 Bit    Registers := 51
                      28 Bit    Registers := 1
```

```
                        5 Bit    Registers := 3
                        3 Bit    Registers := 1
                        2 Bit    Registers := 3
                        1 Bit    Registers := 13
+---RAMs :
                       8K Bit  (256 X 32 bit)          RAMs := 1
+---Muxes :
         257 Input   32 Bit        Muxes := 1
           2 Input   32 Bit        Muxes := 39
           4 Input   32 Bit        Muxes := 6
           4 Input   16 Bit        Muxes := 1
           4 Input    8 Bit        Muxes := 1
           2 Input    8 Bit        Muxes := 1
           2 Input    5 Bit        Muxes := 3
           2 Input    3 Bit        Muxes := 1
           2 Input    2 Bit        Muxes := 1
           2 Input    1 Bit        Muxes := 33
-------------------------------------------------------------------------------
Finished RTL Component Statistics
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
Start Part Resource Summary
-------------------------------------------------------------------------------
Part Resources:
DSPs: 90 (col length:60)
BRAMs: 100 (col length: RAMB18 60 RAMB36 30)
-------------------------------------------------------------------------------
Finished Part Resource Summary
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
Start Cross Boundary and Area Optimization
-------------------------------------------------------------------------------
WARNING: [Synth 8-7080] Parallel synthesis criteria is not met
INFO: [Synth 8-6851] RAM (u_microprocessor0/u_data_memory/u_memory/mem_reg) has partial Byte Wide
Write Enable pattern, however no output register found in fanout of RAM. Recommended to use
supported Byte Wide Write Enable template.
-------------------------------------------------------------------------------
Finished Cross Boundary and Area Optimization : Time (s): cpu = 00:01:18 ; elapsed = 00:01:25 .
Memory (MB): peak = 1254.867 ; gain = 692.570
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
Start ROM, RAM, DSP, Shift Register and Retiming Reporting
-------------------------------------------------------------------------------

Distributed RAM: Preliminary Mapping Report (see note below)
+------------+-----------------------------------------------------+-----------+--------------------
--+-----------------+
|Module Name | RTL Object                                          | Inference | Size (Depth x
Width) | Primitives      |
+------------+-----------------------------------------------------+-----------+--------------------
--+-----------------+
|basys3_top  | u_microprocessor0/u_data_memory/u_memory/mem_reg | Implied   | 256 x 32
| RAM256X1S x 32  |
+------------+-----------------------------------------------------+-----------+--------------------
--+-----------------+

Note: The table above is a preliminary report that shows the Distributed RAMs at the current
stage of the synthesis flow. Some Distributed RAMs may be reimplemented as non Distributed RAM
primitives later in the synthesis flow. Multiple instantiated RAMs are reported only once.
-------------------------------------------------------------------------------
Finished ROM, RAM, DSP, Shift Register and Retiming Reporting
-------------------------------------------------------------------------------
```

```
-----------------------------------------------------------------------------
Start Applying XDC Timing Constraints
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Applying XDC Timing Constraints : Time (s): cpu = 00:01:36 ; elapsed = 00:01:44 . Memory
(MB): peak = 1353.062 ; gain = 790.766
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Timing Optimization
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Timing Optimization : Time (s): cpu = 00:01:39 ; elapsed = 00:01:47 . Memory (MB): peak
= 1376.371 ; gain = 814.074
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start ROM, RAM, DSP, Shift Register and Retiming Reporting
-----------------------------------------------------------------------------

Distributed RAM: Final Mapping Report
+------------+-----------------------------------------------------+-----------+--------------------
--+-----------------+
|Module Name | RTL Object                                          | Inference | Size (Depth x
Width) | Primitives      |
+------------+-----------------------------------------------------+-----------+--------------------
--+-----------------+
|basys3_top  | u_microprocessor0/u_data_memory/u_memory/mem_reg | Implied   | 256 x 32
| RAM256X1S x 32  |
+------------+-----------------------------------------------------+-----------+--------------------
--+-----------------+

-----------------------------------------------------------------------------
Finished ROM, RAM, DSP, Shift Register and Retiming Reporting
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Technology Mapping
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Technology Mapping : Time (s): cpu = 00:01:42 ; elapsed = 00:01:51 . Memory (MB): peak =
1419.094 ; gain = 856.797
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start IO Insertion
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Flattening Before IO Insertion
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Flattening Before IO Insertion
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Final Netlist Cleanup
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Final Netlist Cleanup
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished IO Insertion : Time (s): cpu = 00:01:54 ; elapsed = 00:02:03 . Memory (MB): peak =
1581.066 ; gain = 1018.770
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Renaming Generated Instances
-----------------------------------------------------------------------------
```

```
-----------------------------------------------------------------------------
Finished Renaming Generated Instances : Time (s): cpu = 00:01:54 ; elapsed = 00:02:03 . Memory
(MB): peak = 1581.066 ; gain = 1018.770
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Rebuilding User Hierarchy
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Rebuilding User Hierarchy : Time (s): cpu = 00:01:55 ; elapsed = 00:02:04 . Memory (MB):
peak = 1581.066 ; gain = 1018.770
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Renaming Generated Ports
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Renaming Generated Ports : Time (s): cpu = 00:01:55 ; elapsed = 00:02:04 . Memory (MB):
peak = 1581.066 ; gain = 1018.770
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Handling Custom Attributes
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Handling Custom Attributes : Time (s): cpu = 00:01:55 ; elapsed = 00:02:05 . Memory
(MB): peak = 1581.066 ; gain = 1018.770
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Renaming Generated Nets
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Finished Renaming Generated Nets : Time (s): cpu = 00:01:55 ; elapsed = 00:02:05 . Memory (MB):
peak = 1581.066 ; gain = 1018.770
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
Start Writing Synthesis Report
-----------------------------------------------------------------------------

Report BlackBoxes:
+-+--------------+----------+
| |BlackBox name |Instances |
+-+--------------+----------+
+-+--------------+----------+

Report Cell Usage:
+------+----------+------+
|      |Cell      |Count |
+------+----------+------+
|1     |BUFG      |     2|
|2     |CARRY4    |    46|
|3     |LUT1      |     4|
|4     |LUT2      |   224|
|5     |LUT3      |    49|
|6     |LUT4      |   159|
|7     |LUT5      |   302|
|8     |LUT6      |   920|
|9     |MUXF7     |   256|
|10    |MUXF8     |   128|
|11    |RAM256X1S |    32|
|12    |FDCE      |  1412|
|13    |FDPE      |     2|
|14    |FDRE      |   104|
|15    |IBUF      |     2|
|16    |OBUF      |    16|
```

```
+------+----------+------+
--------------------------------------------------------------------------------
Finished Writing Synthesis Report : Time (s): cpu = 00:01:55 ; elapsed = 00:02:05 . Memory (MB):
peak = 1581.066 ; gain = 1018.770
--------------------------------------------------------------------------------
Synthesis finished with 0 errors, 0 critical warnings and 113 warnings.
Synthesis Optimization Runtime : Time (s): cpu = 00:01:28 ; elapsed = 00:01:58 . Memory (MB):
peak = 1581.066 ; gain = 930.133
Synthesis Optimization Complete : Time (s): cpu = 00:01:56 ; elapsed = 00:02:05 . Memory (MB):
peak = 1581.066 ; gain = 1018.770
INFO: [Project 1-571] Translating synthesized netlist
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.140 . Memory (MB): peak =
1586.223 ; gain = 0.000
INFO: [Netlist 29-17] Analyzing 462 Unisim elements for replacement
INFO: [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds
INFO: [Project 1-570] Preparing netlist for logic optimization
INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.002 . Memory (MB): peak =
1593.914 ; gain = 0.000
INFO: [Project 1-111] Unisim Transformation Summary:
  A total of 32 instances were transformed.
  RAM256X1S => RAM256X1S (MUXF7(x2), MUXF8, RAMS64E(x4)): 32 instances

Synth Design complete | Checksum: a2423959
INFO: [Common 17-83] Releasing license: Synthesis
89 Infos, 101 Warnings, 0 Critical Warnings and 0 Errors encountered.
synth_design completed successfully
synth_design: Time (s): cpu = 00:02:08 ; elapsed = 00:02:24 . Memory (MB): peak = 1593.914 ; gain
= 1222.480
Write ShapeDB Complete: Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.027 . Memory (MB): peak =
1593.914 ; gain = 0.000
INFO: [Common 17-1381] The checkpoint
'C:/Users/vigne/RISCV_FinalSubmission/RISCV_FinalSubmission.runs/synth_1/basys3_top.dcp' has been
generated.
INFO: [Vivado 12-24828] Executing command : report_utilization -file
basys3_top_utilization_synth.rpt -pb basys3_top_utilization_synth.pb
INFO: [Common 17-206] Exiting Vivado at Thu Apr  2 03:13:16 2026…
```

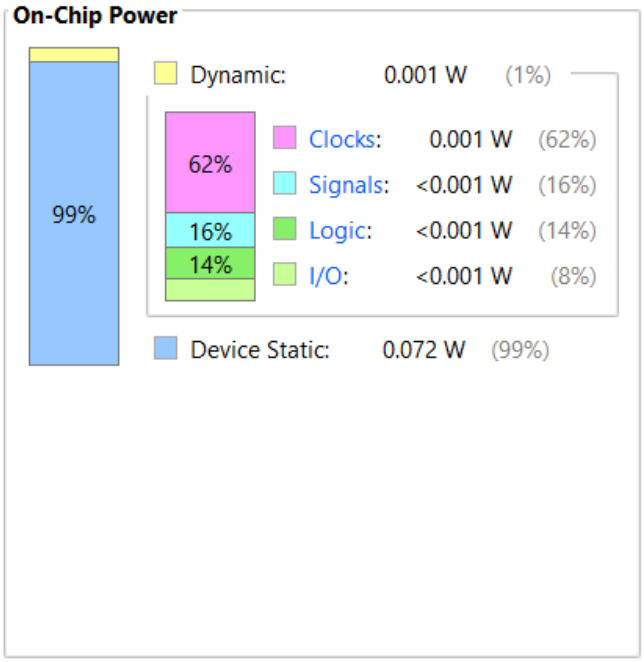
Power Report:

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	**0.073 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**25.4°C**
Thermal Margin:	59.6°C (11.9 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	5.0°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Medium

Launch Power Constraint Advisor to find and fix invalid switching activity




```
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------------------------------------------------------------------------------------------------
-----------------------------------------------------
| Tool Version     : Vivado v.2024.2 (win64) Build 5239630 Fri Nov 08 22:35:27 MST 2024
| Date             : Thu Apr  2 03:16:59 2026
| Host             : Vignesh_Laptop running 64-bit major release  (build 9200)
| Command          : report_power -file basys3_top_power_routed.rpt -pb
basys3_top_power_summary_routed.pb -rpx basys3_top_power_routed.rpx
| Design           : basys3_top
| Device           : xc7a35tcpg236-1
| Design State     : routed
| Grade            : commercial
| Process          : typical
| Characterization : Production
------------------------------------------------------------------------------------------------
-----------------------------------------------------

Power Report

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1. Summary
```

Total On-Chip Power (W)	0.073
Design Power Budget (W)	Unspecified*
Power Budget Margin (W)	NA
Dynamic (W)	0.001
Device Static (W)	0.072
Effective TJA (C/W)	5.0
Max Ambient (C)	84.6
Junction Temperature (C)	25.4
Confidence Level	Medium
Setting File	---
Simulation Activity File	---
Design Nets Matched	NA

* Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>

1.1 On-Chip Components

On-Chip	Power (W)	Used	Available	Utilization (%)
Clocks	<0.001	4	---	---
Slice Logic	<0.001	3858	---	---
LUT as Logic	<0.001	1528	20800	7.35
CARRY4	<0.001	46	8150	0.56
Register	<0.001	1534	41600	3.69
F7/F8 Muxes	<0.001	480	32600	1.47
LUT as Distributed RAM	<0.001	128	9600	1.33
Others	0.000	12	---	---
Signals	<0.001	2570	---	---
I/O	<0.001	18	106	16.98
Static Power	0.072			
Total	0.073			

1.2 Power Supply Summary

Source	Voltage (V)	Total (A)	Dynamic (A)	Static (A)	Powerup (A)	Budget (A)	Margin (A)
Vccint	1.000	0.010	0.001	0.010	NA	Unspecified	NA
Vccaux	1.800	0.013	0.000	0.013	NA	Unspecified	NA
Vcco33	3.300	0.001	0.000	0.001	NA	Unspecified	NA
Vcco25	2.500	0.000	0.000	0.000	NA	Unspecified	NA
Vcco18	1.800	0.000	0.000	0.000	NA	Unspecified	NA
Vcco15	1.500	0.000	0.000	0.000	NA	Unspecified	NA

Vcco135	1.350	0.000	0.000	0.000	NA	Unspecified	NA
Vcco12	1.200	0.000	0.000	0.000	NA	Unspecified	NA
Vccaux_io	1.800	0.000	0.000	0.000	NA	Unspecified	NA
Vccbram	1.000	0.000	0.000	0.000	NA	Unspecified	NA
MGTAVcc	1.000	0.000	0.000	0.000	NA	Unspecified	NA
MGTAVtt	1.200	0.000	0.000	0.000	NA	Unspecified	NA
Vccadc	1.800	0.020	0.000	0.020	NA	Unspecified	NA

1.3 Confidence Level

User Input Data	Confidence	Details	Action
Design implementation state	High	Design is routed	
Clock nodes activity	High	User specified more than 95% of clocks	
I/O nodes activity	Medium	More than 5% of inputs are missing user specification	Provide missing input activity with simulation results or by editing the "By Resource Type -> I/Os" view
Internal nodes activity	Medium	User specified less than 25% of internal nodes	Provide missing internal nodes activity with simulation results or by editing the "By Resource Type" views
Device models	High	Device models are Production	
Overall confidence level	Medium		

2. Settings

2.1 Environment

```
+-----------------------+-------------------------+
| Ambient Temp (C)      | 25.0                    |
| ThetaJA (C/W)         | 5.0                     |
| Airflow (LFM)         | 250                     |
| Heat Sink             | medium (Medium Profile) |
| ThetaSA (C/W)         | 4.6                     |
| Board Selection       | medium (10"x10")        |
| # of Board Layers     | 12to15 (12 to 15 Layers) |
| Board Temperature (C) | 25.0                    |
+-----------------------+-------------------------+


2.2 Clock Constraints
---------------------

+---------+-----------+-----------------+
| Clock   | Domain    | Constraint (ns) |
+---------+-----------+-----------------+
| clk100  | CLK100MHZ |            10.0 |
| cpu_clk | cpu_clk   |    1000000000.0 |
+---------+-----------+-----------------+


3. Detailed Reports
-------------------

3.1 By Hierarchy
----------------

+------+-----------+
| Name | Power (W) |
+------+-----------+
```

Route Status:

```
Design Route Status
                                            :      # nets :
   ------------------------------------------ : ----------- :
   # of logical nets......................... :        4034 :
       # of nets not needing routing......... :        1459 :
           # of internally routed nets........ :        1459 :
       # of routable nets.................... :        2575 :
           # of fully routed nets............. :        2575 :
       # of nets with routing errors......... :           0 :
   ------------------------------------------ : ----------- :
```

Timing Report:

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	5.307 ns	Worst Hold Slack (WHS):	0.087 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	3981	Total Number of Endpoints:	3981	Total Number of Endpoints:	1663

All user specified timing constraints are met.